MULTI-HORIZON FORECASTING OF CARDIOVASCULAR MEDICATION UTILISATION DURING FLOOD DISASTERS:

A COMPARISON OF XGBOOST, LSTM AND TEMPORAL FUSION TRANSFORMER

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INTRODUCTION

Flood disasters **disrupt** medication demand and **supply.**[1]

Predicting medication demand **helps support** disaster preparedness and inventory planning.[2]

CURRENT GAP

The application of **machine learning** to medication forecasting during flood disasters **remains unexplored.**[2,3]

OBJECTIVE

To compare the forecasting performance of
1) XGBoost,
2) Long Short-Term Memory (LSTM),
3) Temporal Fusion Transformer (TFT)
models for predicting cardiovascular medication utilisation across **7-, 14-, 21-, and 28-day** forecast horizons.

METHODS

1. **Input Data** — Medication Utilisation, Weather, Flood
2. **Data Pre-processing**
3. **Feature Engineering**
   - XGBoost Model
   - LSTM Model
   - TFT Model
4. **Multi-Horizon Forecasting** — 7–28 day horizons
5. **Performance Evaluation** — sMAPE, MASE

Figure 1. Overview of the proposed forecasting framework.

sMAPE: Symmetric Mean Absolute Percentage Error
MASE: Mean Absolute Scaled Error

RESULTS

1 How do models perform during floods?

Models **accurately predicted** medication demand during flood observations.

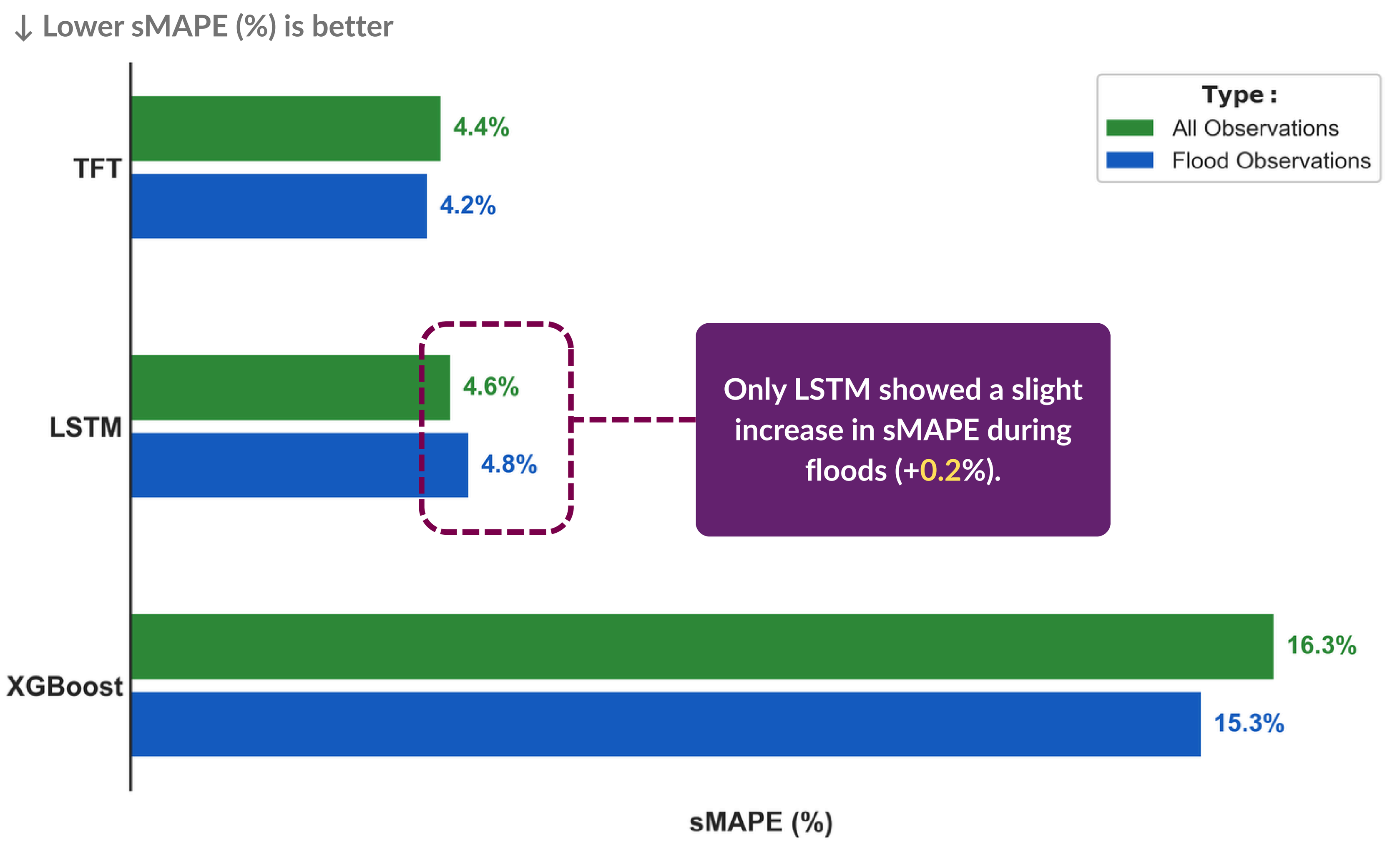

Figure 2. Mean sMAPE across 7- to 28-days horizons.

2 Does forecast horizon matter?

Prediction error **gradually decreased** with longer forecast horizons

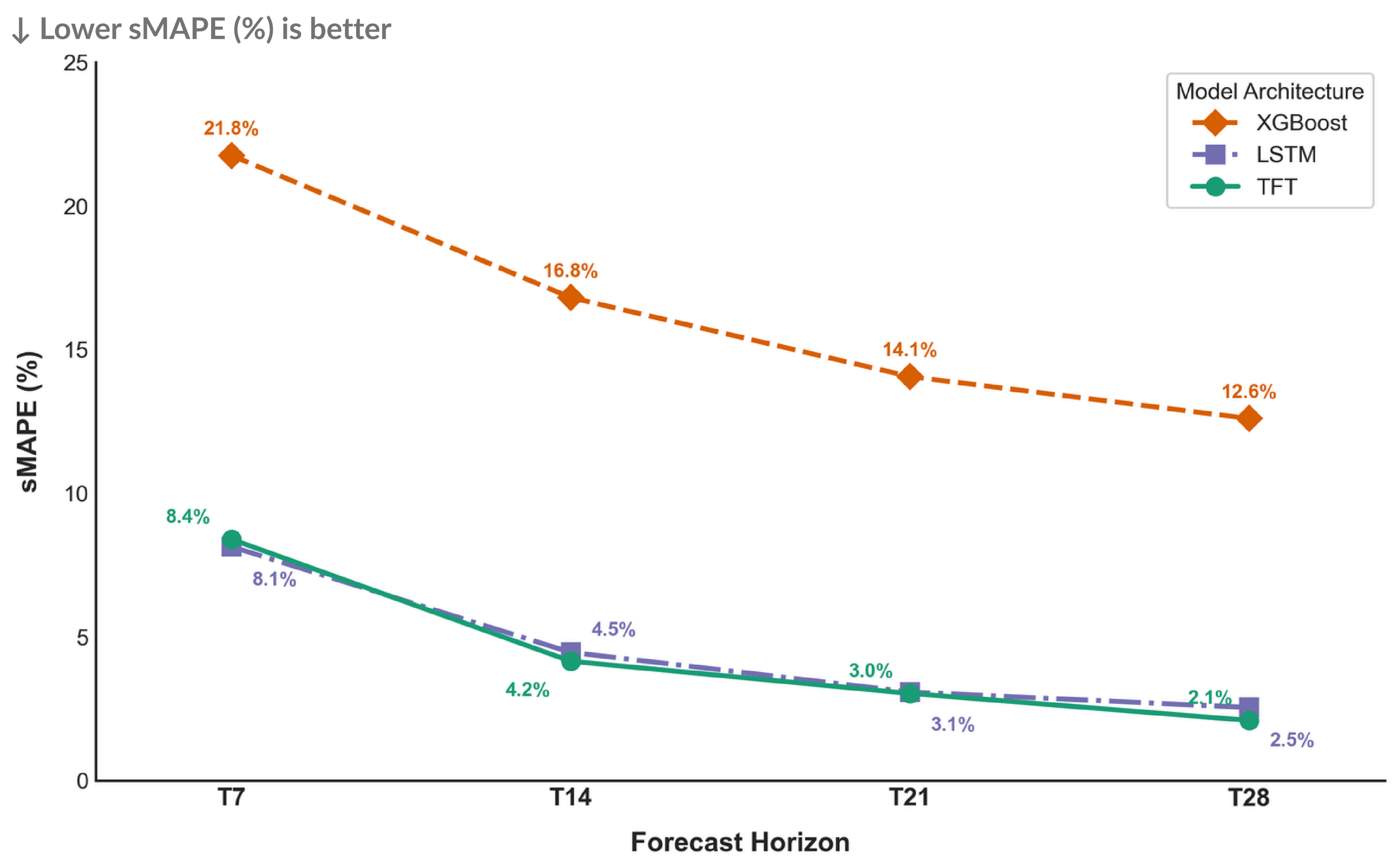

Figure 3. Overall forecasting performance across four horizons.

3 Which model performed most consistently?

TFT was the best-performing model in most forecast horizons.

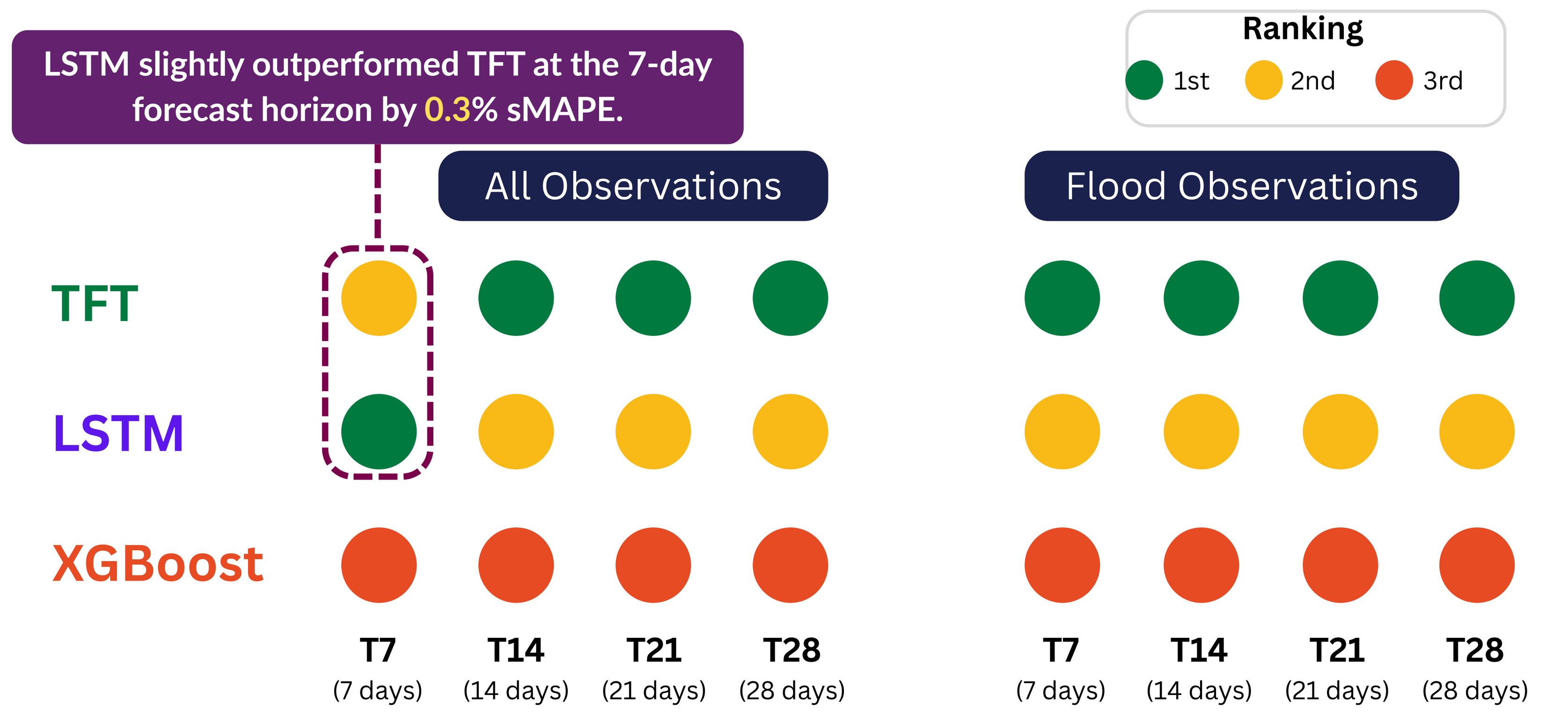

Figure 4. Model ranking across four horizons for overall and flood evaluations.

CONCLUSION

1. Medication demand can be predicted up to **28 days ahead.**
2. **TFT** was the **most promising** forecasting model.
3. Potential to **improve medication planning.**

FUTURE WORK

External validation using an **applicability domain** framework to evaluate model reliability in new settings

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